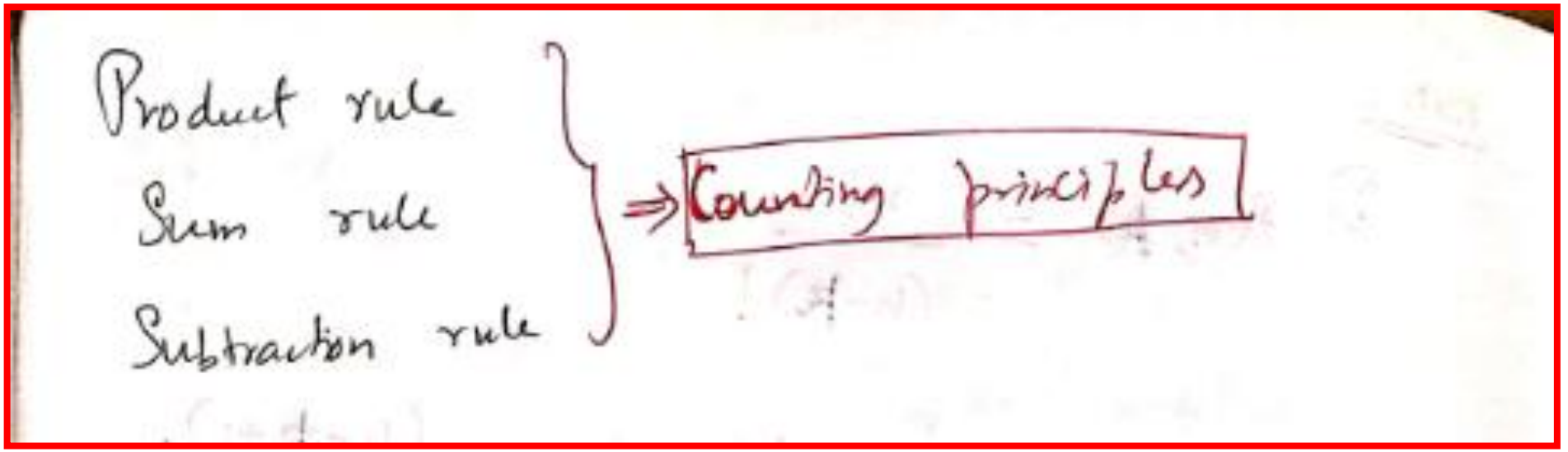
Module 3: Counting Techniques

Basics of counting - Pigeonhole principle - Permutations and combinations – Inclusion – exclusion principle - Recurrence relations - Solving recurrence relations – Generating functions-Solution to recurrence relations.

Counting


Product rule
Sum rule
Subtraction rule
⇒ Counting principles

Product rule:

Suppose that a procedure can be broken down into a sequence of two tasks. If there are n_1-ways to do the first task and for each of those n_1-ways of doing the first task, there are n_2-ways to do the second task, then there are $n_1 n_2$-ways to do the procedure.

Problems:

① How many different bit Strings of length seven are there?

Soln

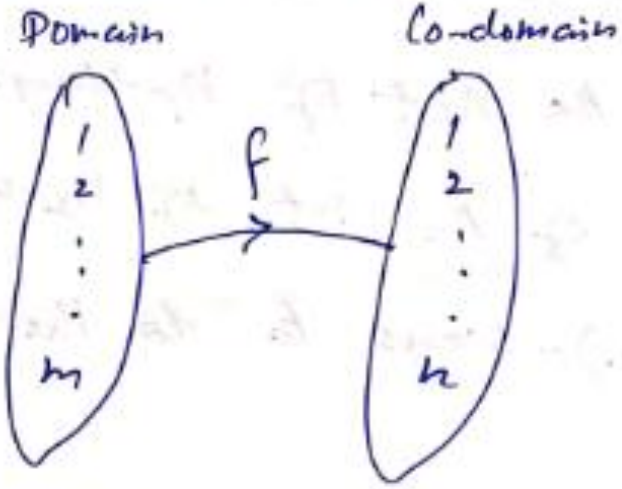
Bit = {0, 1}. and length = 7.

Possibility :- $\underset{1^{st}}{0,1}$ $\underset{2^{nd}}{0,1}$ $\underset{3^{rd}}{0,1}$ $\underset{4^{th}}{0,1}$ $\underset{5^{th}}{0,1}$ $\underset{6^{th}}{0,1}$ $\underset{7^{th}}{0,1}$ ⇒ For each position there is two possibilities.

∴ No. of diff bit Strings = $(2)(2)(2)(2)(2)(2)(2) = 2^7$.
(By product rule)

④. How many functions are there from a set with m-elements to a set with n-elements?

Sol:



$|A| = m$ $\quad$ $|B| = n$

No. of functions from set A to set B $= \underbrace{n \; n \; n \; \ldots \; n}_{m\text{-places.}}$

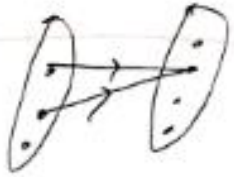
[i.e) Each element of A can be mapped to n-different elements of B].

By product rule,

$$\text{Ans} = (n)(n)(n)\ldots(n)$$

$$= n^m$$

Note:



Function

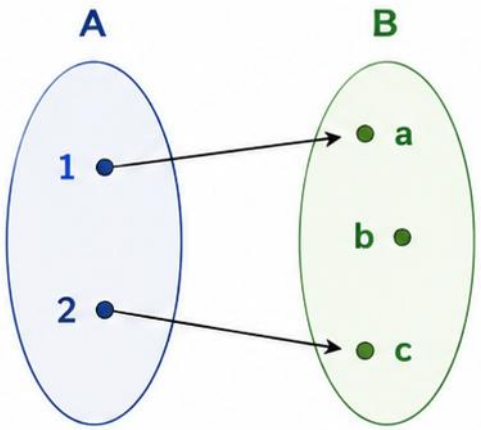


Not a function.

Let $A = \{1, 2\}$ $(|A| = m = 2)$ and $B = \{a, b, c\}$ $(|B| = n = 3)$

1. FUNCTIONS FROM A TO B

In a function, each element of A must be mapped to exactly one element of B.



(One possible function)

Element 1 has 3 choices (a, b, or c). Element 2 also has 3 choices.

↓

Total functions

$= 3 \times 3 = 3^2 = 9$

All 9 functions (arrows show the images):

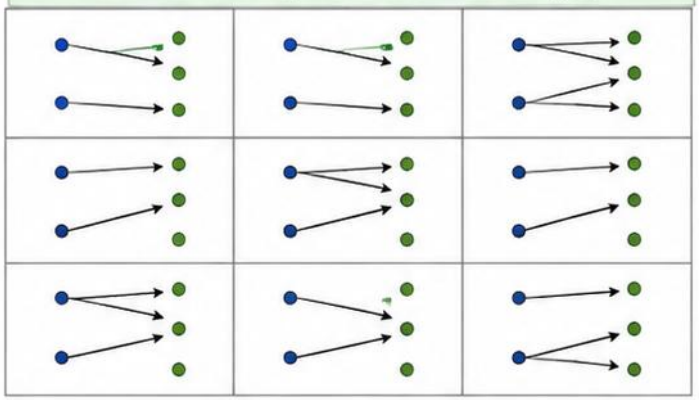


Image of $1 \to a$

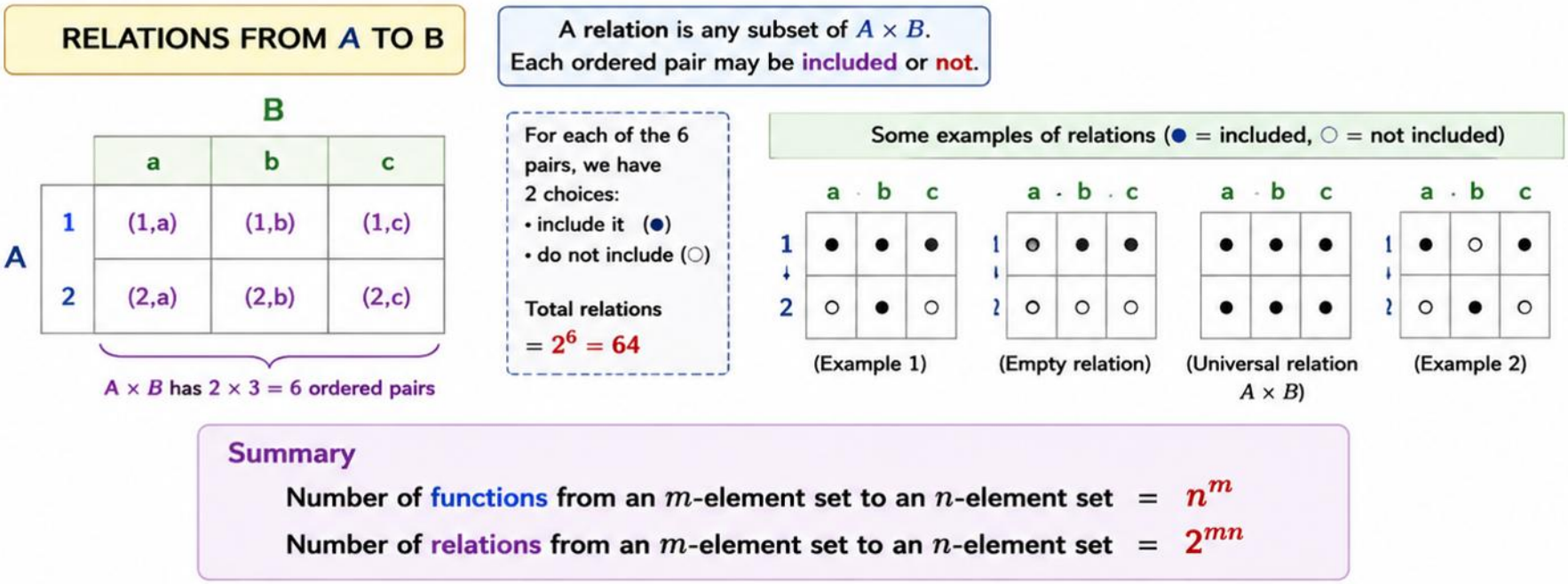
Image of $1 \to b$

Image of $1 \to c$

Image of $2 \to a$ | Image of $2 \to b$ | Image of $2 \to c$

3. How many relations are there from a set A with m elements to a set B with n elements?

Solution: Number of relations from A to B is 2^{mn}.



Sum rule:-

If a task can be done either in one of n_1-ways or in one of n_2 ways, where none of the set of n_1-ways is the same as any of the set of n_2-ways, then there are $(n_1 + n_2)$-ways to do the task.

Problems:-

(1) A Student can choose a computer project from one of three lists. The three lists contain 23, 15 and 19 possible projects, respectively. No projects is on more than one list. How many possible projects are there to choose from?

Sol:-

No. of possible projects to choose } = 23+15+19.

(By Sum rule)

Since no project is on more than one list.

(2) Each user on a computer system has a password, which is 6 to eight characters long, where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there?

Sol:-

Let $P \rightarrow$ Total no. of possible passwords.

$P_6 \rightarrow$ No. of possible passwords of length 6.

$P_7 \rightarrow$ " " " " 7.

$P_8 \rightarrow$ " " " " 8.

Now,

$P_6 \Rightarrow$ _ _ _ _ _ _

Given, Each password must contain at least one digit.

P_6 = [No. of Strings with upper case and digits] − [No. of Strings with no digits].

$= [36\ 36\ 36\ 36\ 36\ 36] - [26\ 26\ 26\ 26\ 26\ 26]$

Note:- [26 + 10 = 36]
Alphabet — ↳ Digits.

$= 36^6 - 26^6 = 1,867,866,560.$

III^{ly} $P_7 = 36^7 - 26^7 = 70,333,353,920.$

and $P_8 = 36^8 - 26^8 = 2,612,282,842,880.$

$\therefore P = P_6 + P_7 + P_8.$

$$\boxed{P = 2,684,483,063,360.}$$

4.

Subtraction rule :- [Principle of inclusion - exclusion].

If a task can be done in either n_1-ways or n_2-ways, then the no. of ways to do the task is $n_1 + n_2$ - (no. of ways to do the task that are common to the two different ways).

Note:

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

No. of ways do select an element in A or in B is $= |A| + |B| - |A \cap B|$.

Problems:

(7) How many bit strings of length eight either start with a 1 bit or end with the two bits 00?.

Sol:- Bit = $\{0,1\}$. and Length = 8.

String start with a 1-bit = $\underline{1}\ \underbrace{\underline{0,1}\ \underline{0,1}\ \underline{0,1}\ \underline{0,1}\ \underline{0,1}\ \underline{0,1}\ \underline{0,1}}_{2^7\text{-ways.}}$

Ending with two bits 00 = $\underbrace{\underline{0,1}\ \underline{0,1}\ \underline{0,1}\ \underline{0,1}\ \underline{0,1}\ \underline{0,1}}_{2^6\text{-ways}}\ \underline{0}\ \underline{0}$

Bit string with 1-bit at the start and ending with two bits 00.

$= \underline{1}\ \underline{0,1}\ \underline{0,1}\ \underline{0,1}\ \underline{0,1}\ \underline{0,1}\ \underline{0}\ \underline{0}$

2^5-ways.

$\therefore$ No. of ways to construct a bit string of length eight that begin with a 1 or that ends with 00-bits

$= 2^7 + 2^6 - 2^5$

$= 160.$ Ans.

Number of *k*-Sequences on an *n*-Set

When the codomain of a sequence S is the set C, we say that S is a ***sequence on*** C. If both k and n are positive integers, then a *k-sequence on an n-set* is a function S from $\{1..k\}$ into some set $X = \{x_1, x_2, \ldots, x_n\}$ with exactly n elements, and we may write S as

$$S = (s_1, s_2, s_3, \ldots, s_k) \text{ where each } s_j \in X$$

We can count these like we did the telephone numbers using the product rule.

There are n possible choices for s_1.
After s_1 is chosen, there are n possible choices for s_2.
After s_1 and s_2 are chosen, there are n possible choices for s_3, and so on.

Since there are n possible choices for each of the k entries in S,

the number of k-sequences on an n-set is $n \times n \times \ldots \times n = \boldsymbol{n^k}$

In particular, the number of k-sequences on $\{0,1\}$ is 2^k

Number of *k*-Permutations on an *n*-Set

A permutation is a sequence without repetitions.

Then a *k-permutation on an n-set* is a function S from $\{1..k\}$ into some set $X = \{x_1, x_2, \ldots, x_n\}$ with exactly n elements that is ***one-to-one***; that is, if $i \neq j$, then $S(i) \neq S(j)$. We may write S as

$$S = (s_1, s_2, s_3, \ldots, s_k) \text{ where the } s_j\text{'s are distinct elements of X.}$$

If $1 <= k <= n$, we can count these like we did sequences using the product rule. There are n possible choices for s_1.

After s_1 is chosen, there are $(n - 1)$ possible choices for s_2.

After s_1 and s_2 are chosen, there are $(n - 2)$ possible choices for s_3.

After s_1 up to s_j have been chosen, there are $(n - j)$ possible choices for s_{j+1}.

After s_1 up to s_{k-1} have been chosen and then there are $(n - [k - 1])$ possible choices for s_k.

Thus, if $1 \leq k \leq n$, the number of k −permutations on an n −set is

$$n \times (n-1) \times (n-2) \times \ldots \times (n-(k-1))$$

n-Factorial:

When $k = n$, each k-permutation uses *all* of the elements of the n-set X exactly once; we'll call these ***full permutations*** of X. The number of full permutations of an n-set is

$$n \times (n-1) \times (n-2) \times \ldots \times (2) \times (1)$$

Number of *k*-Permutations on an *n*-Set

$$P(n,k) = \frac{n!}{(n-k)!}$$

Number of k-Subsets of an n-Set

The complex symbol $\binom{n}{k}$ will denote the number of k-subsets of an n-set assuming $n \geq k \geq 0$.

The number of k-subsets of an n-set assuming

$$C(n,k) = \binom{n}{k} = \frac{n!}{k! \times (n-k)!}$$

Suppose X is any n-set. The empty set is the only subset of X of size 0, and all of X is the only subset of size n, so

$$\binom{n}{0} = 1 \text{ and } \binom{n}{n} = 1$$

There are exactly n subsets of size 1, so

$$\binom{n}{1} = n$$

Note:

- $$\binom{n}{n-k} = \frac{n!}{(n-k)! \times [n-(n-k)]!}$$

- $$\binom{n}{n-k} = \binom{n}{k}$$

- $$\binom{n-1}{k} + \binom{n-1}{k-1} = \binom{n}{k}$$

Problems:

① In how many ways can we select three students from a group of five students to stand in the line for a picture? In how many ways can we arrange all five of these students in a line for a picture?

Sol:-

Note that the order in which we select the students matters.

By the product rule, we have $(5)(4)(3) = 60$

ways to select three students from a group of five students to stand in line for a picture.

Now, the no of ways to arrange all five students in a line for a picture } = (5)(4)(3)(2)(1)

= 120. (= 5!)

Note:

Three students in a line } ⇒ 5-ways 4-ways 3-ways (Since students can't be repeated).

Five students in a line } ⇒ 5-W 4-W 3-W 2-W 1-W

(2) How many ways are there to select a first-prize winner, a second-prize winner, and a third-prize winner from 100 different people who have entered a contest?

Sol:-

No. of ways to select 1st, 2nd & 3rd prize winner from 100 different people $= \underset{1^{st}}{(100)} \underset{2^{nd}}{(99)} \underset{3^{rd}}{(98)}$

$= 970,200.$

$[= P(100, 3)$

$= (100)(100-1)(100-2),$

$= 970,200].$

(13) How many poker hands of five cards can be dealt from a standard deck of 52 cards? Also, how many ways are there to select 47 cards from a standard deck of 52 cards?

Sol:-

No. of poker hands of 5 cards from 52 cards $= C(52, 5)$

$$= \frac{52!}{5!(52-5)!}$$

$$= \frac{52!}{5!\,47!}$$

$$= 2,598,960.$$

No. of ways selecting 47 cards out of 52 cards $\Big\}$ $= C(52, 47)$

$$= \frac{52!}{47!\,(52-47)!}$$

$$= \frac{52!}{47!\,5!}$$

$$= 2,598,960.$$

Note:-

$$C(n, k) = C(n, n-k)$$

(4). Suppose that there are 9 faculty members in the mathematics department and 11 in the computer science department. How many ways are there to select a committee to develop a discrete mathematics course at a college if the committee is to consist of three faculty members from the mathematics dept and four from the computer science dept?

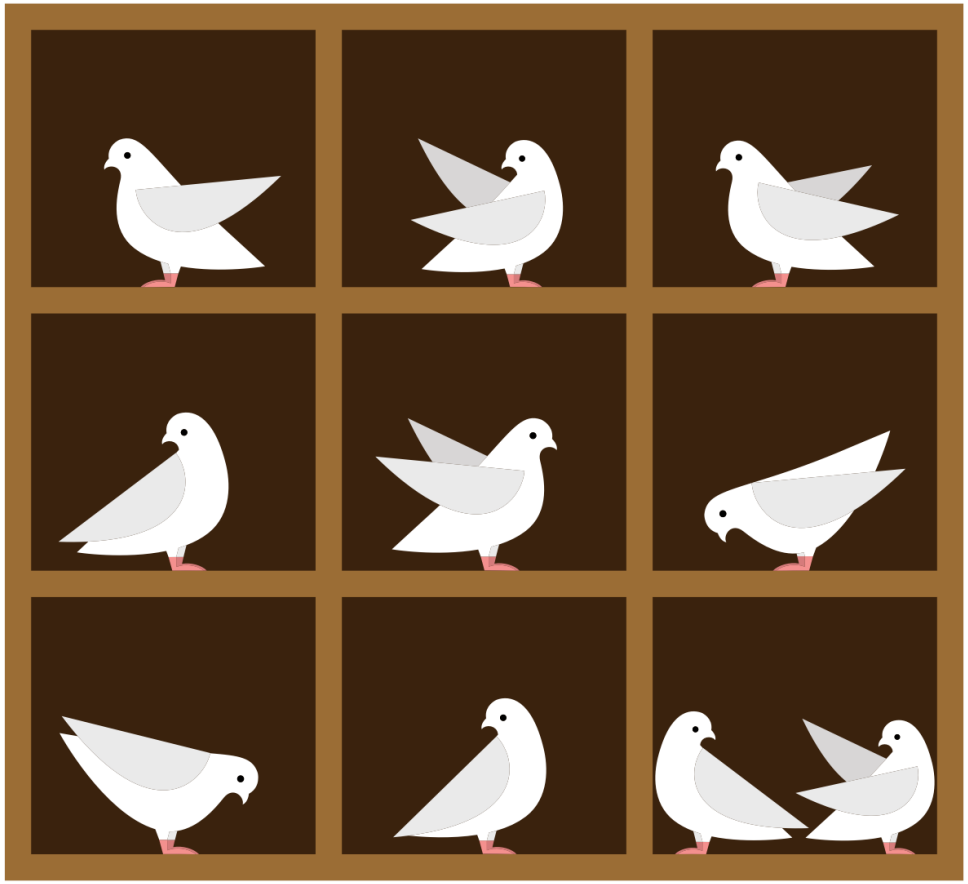
Sol'n

By the product rule and by the combination theory, we have,

No. of ways to select a committee to develop a discrete mathematics course $\Big\}$ $= C(9,3) \cdot C(11,4)$

$= \left[\frac{9!}{3!6!}\right]\left[\frac{11!}{4!7!}\right] = 27,720.$

Pigeonhole Principle



Counting Argument

The Pigeonhole Principle

- The **pigeonhole principle** states that "if P pigeons flies into H pigeons holes then at least one pigeon hole contains at least $\left\lceil \frac{P}{H} \right\rceil$ pigeons, where $P > H$".

- In other words, if N objects are placed into k −boxes, then there is at least one box containing at least $\left\lceil \frac{N}{\mathrm{k}} \right\rceil$ objects, when $N > k$.

Eg:- (1) Among any group of 367 people, there must be at least two with the same birthday, because there are only 366 possible birthdays. (Including leap year)

(2) How many Students must be in a class to guarantee that at least two Students receive the same score on the final exam, if the exam is graded on a scale from 0 to 100 points?

Ans:

No. of possible scores = 101.

By pigeonhole principle, among any 102 Students there must be at least two Students with the same score.

Floor and Roof Functions

Roof function of x $\lceil x \rceil$: Nearest integer greater than x

Eg: $\lceil 2.32 \rceil = 3$

Floor function $\lfloor x \rfloor$: Nearest integer lesser than x

Eg: $\lfloor 2.32 \rfloor = 2$

Problems:

① What is the minimum no. of students required in a discrete mathematics class to be sure that at least six will receive the same grade, if there are five possible grades, A, B, C, D and F?

Sol:-

Let N be the minimum no. of students required. Also we have five possible grades.

Given:- $\lceil N/5 \rceil = 6$

$1 \leq r \leq 5$

Now, $N = 5(5) + r$,

Since N is minimum, we must have $r = 1$.

$\therefore N = 5(5) + 1 = 26$.

$\therefore$ There has to a minimum of 26 students required in a discrete mathematics class to be sure that at least two will receive the same grade.

(2) What is the least no. of area codes needed to guarantee that the 25 million phones in a State can be assigned distinct 10-digit telephone numbers? [Note: NXX-NXX-XXXX is the telephone number.
(Area code)
where, N → Digits from 2 to 9.
X → Digits from 0 to 9.].

Sol:-

Phone number: NXX-NXX-XXXX

Later we can see this.

There are how many possible way?

NXX-XXXX ⇒ Our main concern is the last 7-digits of the phone number

W.K.T N → Digits from 2 to 9.

∴ There are 8 possible ways we can choose the values of N.

X → Digits from 0 to 9.

∴ There are 10 possible ways we can choose the values of X.

By the "product rule for counting", we have

No. of ways $\Rightarrow$ [8] [10] [10] [10] [10] [10] [10]

N X X X X X X

No. of distinct phone numbers of the form NXX-XXXX $= (8)(10)(10)(10)(10)(10)(10)$

$= 8{,}000{,}000.$

$= 8$ million.

By the "generalized pigeonhole principle"

No. of identical phone numbers $\geq \left\lceil \frac{25{,}000{,}000}{8{,}000{,}000} \right\rceil = 4$

(25,000,000: Total no. of phones; 8,000,000: Total no. of different phone numbers)

Now to avoid that at least four identical from numbers we need at least four area codes.

Hence, at least four area codes are required to ensure that all 10-digit numbers are different.

(3) Suppose that 27 pigeons fly into a coop (an apartment house for pigeons) with 5 pigeonholes (apartments). What is the maximum number M that could fly into any single pigeonhole? It is clear that $M \leq 27$. But is there a lower bound on M?.

Sol:-

Let the holes be numbered from 1 to 5.

Let x_i be the no of pigeon that flies into the hole number i.

Then, $x_1 + x_2 + x_3 + x_4 + x_5 = 27$.

W.K.T, $x_i \leq M$, $M \rightarrow$ maximum.

$$\Rightarrow 27 = x_1 + x_2 + x_3 + x_4 + x_5$$

$$\leq M + M + M + M + M$$

$27 \leq 5M \Rightarrow 5M \geq 27.$

$M \geq \frac{27}{5}$

$M \geq \left\lceil \frac{27}{5} \right\rceil = 6.$

∴ The minimum no. of pigeon that flies into the pigeonhole is 6.

Recurrence Relations

- An equation that expresses a_n namely the general term of the sequence $\{a_n\}$ in terms of one or more of the previous terms of the sequence, namely $a_0, a_1, a_2, \ldots, a_{n-1}$, for all integers n with $n \geq n_0$, where n_0 is a non-negative integer is called a ***recurrence relation*** **for** $\{a_n\}$ **or a** ***difference equation.***

Example

- Consider a geometric progression 4, 12, 36, 108, . . . The common ratio of which is 3.
- If $\{a_n\}$ represents this infinite sequence,

we see that $\frac{a_{n+1}}{a_n} = 3 \Longrightarrow \boxed{a_{n+1} = 3a_n,\ \ n \geq 0}$ is the recurrence relation corresponding to the geometric sequence.

- **However, the above recurrence relation does not represent a unique geometric sequence.** Since the sequence 5, 15, 45, 135, . . . also satisfies the above relation.

- In order that the recurrence relation recurrence relation

$a_{n+1} = 3a_n, n \geq 0$ may represent a unique sequence, we should know one of the terms of the sequence, say, $a_0 = 4$.

- The value $a_0 = 4$ is called the ***initial condition.***
- Consider a Fibonacci sequence 0, 1, 1, 2, 3, 5, 8, 13, 21, . . . , which can be represented by the recurrence relation $F_{n+2} = F_{n+1} + F_n$, where $n \geq 0$ and $F_0 = 0, F_1 = 1$.

Recurrence Relation - General Form

- A recurrence relation of the form

$$c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} = f(n)$$

is called a **recurrence relation of degree k** with constant coefficients, where $c_0, c_1, c_2, \ldots, c_k$ are real numbers and $c_k \neq 0$.

- The **recurrence relation is called linear** because each a_r is raised to the power one and there are no products such as $a_r . a_s$.
- The **degree** is the difference between the greatest and least subscripts of the members of the sequence occurring in the recurrence relation.

A recurrence relation defines the terms of a sequence in terms of earlier terms. A linear recurrence relation of order k can be written as

$$a_n + c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} = f(n),$$

where $c_1, \ldots, c_k$ are constants and $f(n)$ is a known function.

If

$$f(n) = 0,$$

the recurrence is called **homogeneous**. If

$$f(n) \neq 0,$$

it is called **non-homogeneous**.

For a non-homogeneous linear recurrence, two important methods are:

1. the **Complementary Function + Particular Integral (C.F. + P.I.)** method, usually implemented using the method of undetermined coefficients;

2. the **Generating Function** method.

Method 1:

Complementary Function

+

Particular Integral (Trial solution)

Note: Problems will not be asked using Method 1. This method is discussed only to provide motivation and intuition for Method 2.

Method 1: Complementary Function + Particular Integral

Summar Table for Choosing C.F.

Roots of characteristic equation	Complementary Function (C.F.)
λ_1, λ_2 distinct and real	$C_1\lambda_1^n + C_2\lambda_2^n$
$\lambda_1 = \lambda_2 = \lambda$ repeated real root	$(C_1 + C_2 n)\lambda^n$
$\lambda_{1,2} = a \pm ib$	$\rho^n \left[C_1 \cos(n\theta) + C_2 \sin(n\theta)\right]$
λ has multiplicity s	$(C_0 + C_1 n + \cdots + C_{s-1} n^{s-1})\lambda^n$

where, for the complex-root case,

$$\rho = \sqrt{a^2 + b^2}, \qquad \cos\theta = \frac{a}{\rho}, \qquad \sin\theta = \frac{b}{\rho}.$$

Summary Table for Choosing the P.I.

RHS $f(n)$	**Basic trial P.I.**
$P_m(n)$	$Q_m(n)$
λ^n	$C\lambda^n$
$\cos(\theta n)$ or $\sin(\theta n)$	$C_1\cos(\theta n) + C_2\sin(\theta n)$
$P_m(n)\lambda^n$	$\lambda^n Q_m(n)$
$\lambda^n\cos(\theta n)$ or $\lambda^n\sin(\theta n)$	$\lambda^n[C_1\cos(\theta n) + C_2\sin(\theta n)]$
$P_m(n)\lambda^n\cos(\theta n)$ or $P_m(n)\lambda^n\sin(\theta n)$	$\lambda^n[Q_m(n)\cos(\theta n) + R_m(n)\sin(\theta n)]$

Here P_m, Q_m, R_m denote polynomials of degree m.
After choosing the basic trial, apply the overlap rule:

$$\boxed{\text{P.I.} = n^s(\text{basic trial})}$$

when the corresponding characteristic root has multiplicity s.

Procedure :-

(1) Get the roots λ_1 and λ_2 by taking R.H.S = 0.

(2) Find C.F, y_h^c

(3) Find P.I using method of undetermined coefficient $y_h^{P.I}$.

(4) General solution

$$y_h = C.F + P.I.$$

Method 2: Generating Functions

Note: Only Method 2 is in our syllabus. Recurrence problems must be solved using the method of generating functions.

Solution of recurrence relations using Generating Functions

The generating function of a sequence $a_0, a_1, a_2, \ldots$, is the expression

$$G(x) = a_0 + a_1 x + a_2 x^2 + \cdots = \sum_{n=0}^{\infty} a_n x^n$$

$$G(x) = \sum_{n=0}^{\infty} a_n x^n$$

Undetermined Coefficients vs Generating Functions

Feature	Undetermined Coefficients	Generating Functions
Basic idea	Guess a particular solution	Encode the sequence in a power series
Guess required?	Usually yes	No
Initial conditions	Applied after obtaining general solution	Incorporated naturally while forming $G(x)$
Non-homogeneous terms	May require complicated guesses	Handled systematically if their generating functions are known
Best suited for	Many standard linear recurrences	A broad class of sequence and recurrence problems
Main challenge	Choosing the correct particular form	Algebraic manipulation and coefficient extraction

Useful Power-Series Expansions

1. The generating function for the sequence 1,1,1,1, ... is given by

$$G(x) = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}, \qquad \text{for } |x| < 1$$

2. The generating function for the sequence 1,2,3,4, ... is given by

$$G(x) = \sum_{n=0}^{\infty} (n+1)x^n = 1 + 2x + 3x^2 + \cdots$$

$$G(x) = \frac{1}{(1-x)^2}, \qquad \text{for } |x| < 1$$

3. The generating function for the sequence $1, a, a^2, a^3, \ldots$ is given by

$$G(x) = \sum_{n=0}^{\infty} (ax)^n = 1 + ax + a^2x^2 + \cdots$$

$$G(x) = \frac{1}{1-ax}, \text{ for } |ax| < 1.$$

Solving Procedure

Consider $a_n + c_1\, a_{n-1} + c_1 a_{n-1} + \cdots + c_n a_{n-k} = f(n)$

Step 1: Define

$$G(x) = \sum_{n=0}^{\infty} a_n x^n$$

Step 2: Multiply the recurrence relation by x^n.

Step 3: Sum both sides over the appropriate values of n.

Step 4: Rewrite every summation in terms of $G(x)$.

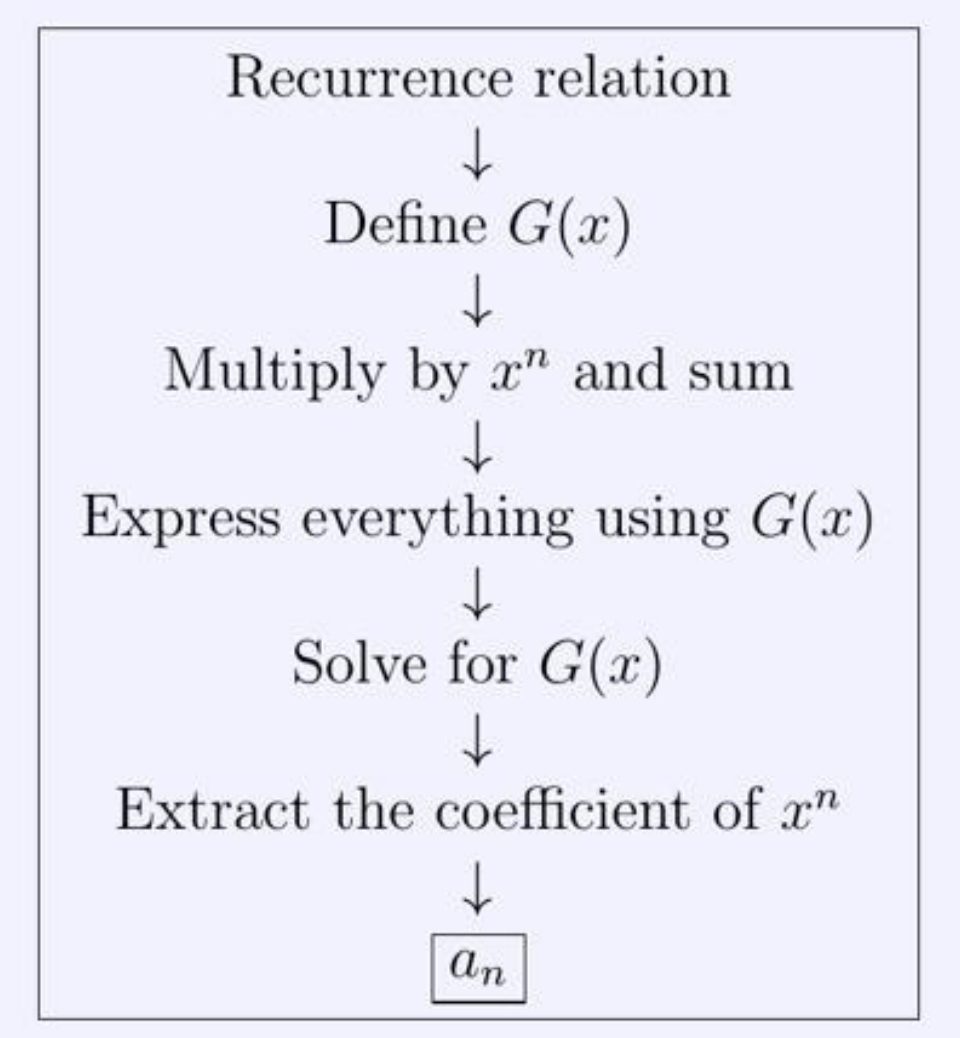
Step 5: Use the initial conditions to simplify the expression.

Step 6: Solve the resulting algebraic equation for $G(x)$.

Step 7: Expand $G(x)$, use partial fractions, or otherwise extract the coefficient of x^n.

Step 8: a_n = coefficient of x^n in the expansion of $G(x)$

Key Idea


Recurrence relation
↓
Define $G(x)$
↓
Multiply by x^n and sum
↓
Express everything using $G(x)$
↓
Solve for $G(x)$
↓
Extract the coefficient of x^n
↓
a_n

Problems:

1. Use the method of generating function to solve the recurrence relation $a_n = 3a_{n-1} + 1;\ n \geq 1$, given that $a_0 = 1$.

2. Use the method of generating function to solve the recurrence relation $a_n = 4a_{n-1} - 4a_{n-2} + 4^n;\ n \geq 2$, given that $a_0 = 2$ and $a_1 = 8$.

3. Use the method of generating function to solve the recurrence relation $a_{n+1} - 8a_n + 16a_{n-1} = 4^n;\ n \geq 1$, given that $a_0 = 1$ and $a_1 = 8$.

4. Use the method of generating function to solve the recurrence relation $a_n = 4a_{n-1} + 3n.2^n;\ n \geq 1,$ given that $a_0 = 4$.

Problem:

1. Use the method of generating function to solve the recurrence relation $a_n = 3a_{n-1} + 1;\ n \geq 1$, given that $a_0 = 1$.

Sol:-

Given: $a_n = 3a_{n-1} + 1$, $n \geq 1$ and $a_0 = 1$

$$\sum_{n=1}^{\infty} a_n x^n = 3 \sum_{n=1}^{\infty} a_{n-1} x^n + \sum_{n=1}^{\infty} x^n \quad \text{---(1)}$$

W.K.T, Generating function, $G(x) = \sum_{n=0}^{\infty} a_n x^n$

Now, $\sum_{n=1}^{\infty} a_n x^n = G(x) - a_0$ — (2)

$$G(x) = \sum_{n=0}^{\infty} a_n x^n$$

$$= a_0 x^0 + \sum_{n=1}^{\infty} a_n x^n$$

$$\sum_{n=1}^{\infty} a_n x^n = G(x) - a_0$$

Now, $3\sum_{n=1}^{\infty} a_{n-1} x^n = 3x \sum_{n=1}^{\infty} a_{n-1} x^{n-1}$

$$= 3x \sum_{n=0}^{\infty} a_n x^n \quad \text{— (3)}$$

Now, $\sum_{n=1}^{\infty} x^n = x \sum_{n=1}^{\infty} x^{n-1} = x \sum_{n=0}^{\infty} x^n$

$$\sum_{n=1}^{\infty} x^n = x \sum_{n=0}^{\infty} x^n \quad \text{---} (4)$$

Substitute (2), (3) & (4) in (1).

$$(1) \Rightarrow \sum_{n=1}^{\infty} a_n x^n = 3x \sum_{n=1}^{\infty} a_{n-1} x^n + \sum_{n=1}^{\infty} x^n$$

$$G(x) = \sum_{n=0}^{\infty} a_n x^n$$

$$G(x) - a_0 = 3x \sum_{n=0}^{\infty} a_n x^n + x \sum_{n=0}^{\infty} x^n$$

$$G(x) - a_0 = 3x\, G(x) + x\left(\frac{1}{1-x}\right)$$

$$G_1(x) - 3x\,G_1(x) = \frac{x}{1-x} + a_0$$

$$(G_1(x))(1-3x) = \frac{x}{1-x} + 1 \qquad [\because a_0 = 1]$$

$$G_1(x) = \frac{1}{(1-x)(1-3x)} \quad \text{—— (5)}$$

Now,

$$\frac{1}{(1-x)(1-3x)} = \frac{A}{1-x} + \frac{B}{1-3x}$$

After Solving $\Rightarrow$ $A = -1/2$, $B = 3/2$

Now, $G(x) = \frac{-1/2}{1-x} + \frac{3/2}{1-3x}$

$= -\frac{1}{2}(1-x)^{-1} + \frac{3}{2}(1-3x)^{-1}$

$G(x) = -\frac{1}{2}\left(\sum_{n=0}^{\infty} x^n\right) + \frac{3}{2}\left(\sum_{n=0}^{\infty} (3x)^n\right)$

Now, a_n = Co.eff. of x^n in $G(x)$

$= -\frac{1}{2} + \frac{3}{2}(3)^n \Rightarrow \boxed{a_n = \frac{1}{2}(3^{n+1} - 1)}$

α

Formula:

General form: $(1-x)^{-n}$

Terms: $T_{r+1} = (n+r-1)C_r \cdot x^r$

Eg: $(1-2x)^{-2} \Rightarrow x = 2x$ and $n = 2$

$r=0,\ T_1 = (2+0-1)C_0 (2x)^0 = 1$

$r=1,\ T_2 = (2+1-1)C_1 (2x)^1 = 2(2x)^1$

$r=2,\ T_3 = (2+2-1)C_2 (2x)^2 = 3(2x)^2$

$\vdots$

Now,

$$(1-2x)^{-2} = 1 + 2.(2x)^1 + 3(2x)^2 + \cdots$$

2. Use the method of generating function to solve the recurrence relation $a_n = 4a_{n-1} - 4a_{n-2} + 4^n$; $n \geq 2$, given that $a_0 = 2$ and $a_1 = 8$.

Sol:-

Given: $a_n = 4a_{n-1} - 4a_{n-2} + 4^n$, $n \geq 2$

Initial conditions: $a_0 = 2$, $a_1 = 8$

$$\sum_{n=2}^{\infty} a_n x^n = 4\sum_{n=2}^{\infty} a_{n-1}x^n - 4\sum_{n=2}^{\infty} a_{n-2}x^n + \sum_{n=2}^{\infty} 4^n x^n \quad \text{--- (1)}$$

Now, $\sum_{n=2}^{\infty} a_n x^n = h(x) - a_0 - a_1 x$

$= h(x) - 2 - 8x$

W.K.T, $h(x) = \sum_{n=0}^{\infty} a_n x^n$

$h(x) = a_0 + a_1 x + \sum_{n=2}^{\infty} a_n x^n$

$\sum_{n=2}^{\infty} a_n x^n = h(x) - a_0 - a_1 x$

Now,

$4 \sum_{n=2}^{\infty} a_{n-1} x^n = 4x \sum_{n=2}^{\infty} a_{n-1} x^{n-1}$

$= 4x \left[\sum_{n=2}^{\infty} a_{n-1} x^{n-1} + a_0 - a_0 \right]$

$= 4x \left[\sum_{n=1}^{\infty} a_{n-1} x^{n-1} - a_0 \right] = 4x \left[\sum_{n=0}^{\infty} a_n x^n - a_0 \right]$

$$4\sum_{n=2}^{\infty} a_{n-1}x^n = 4x\left[G(x) - a_0\right] = 4x\,G(x) - 4x.a_0.$$

$$= 4x\,G(x) - 8x \quad [\because a_0 = 2].$$

Now,

$$4\sum_{n=2}^{\infty} a_{n-2}x^n = 4x^2 \sum_{n=2}^{\infty} a_{n-2}x^{n-2}$$

$$= 4x^2 \sum_{n=0}^{\infty} a_n x^n$$

$$= 4x^2.G(x)$$

Now, $\sum_{n=2}^{\infty} 4^n x^n = \sum_{n=2}^{\infty} (4x)^n$

$= \sum_{n=0}^{\infty} (4x)^n - 1 - 4x$

W.K.T,

$\sum_{n=0}^{\infty} (ax)^n = 1 + (ax)^1 + (ax)^2 + (ax)^3 + \cdots$

$\sum_{n=0}^{\infty} (4x)^n = 1 + (4x) + (4x)^2 + (4x)^3 + \cdots$

$\sum_{n=0}^{\infty} (4x)^n - 1 - 4x = \sum_{n=2}^{\infty} (4x)^n$

$$\sum_{n=2}^{\infty} a_n x^n = 4\sum_{n=2}^{\infty} a_{n-1}x^n - 4\sum_{n=2}^{\infty} a_{n-2}x^n + \sum_{n=2}^{\infty} 4^n x^n$$

Initial Cond

(1)

$$G(x) - 2 - 8x = \left[4x\,G(x) - 8x\right] - \left[4x^2 G(x)\right] + \left[\sum_{n=0}^{\infty} (4x)^n - 1 - 4x\right]$$

$$G(x)\left[1 - 4x + 4x^2\right] = 8x + 2 - 8x + \sum_{n=0}^{\infty} (4x)^n - 1 - 4x$$

$$\left(G(x)\right)(1-2x)^2 = \sum_{n=0}^{\infty} (4x)^n - 4x + 1$$

$$(G(x)).(1-2x)^2 = \frac{1}{1-4x} - 4x + 1$$

$$= \frac{1}{1-4x} + 1 - 4x.$$

$$(1-2x)^2 G(x) = \frac{1 + (1-4x)^2}{1-4x}.$$

$$G(x) = \frac{1 + (1-4x)^2}{(1-2x)^2 (1-4x)}$$

Apply Partial fraction.

W.k.T

$$\sum_{n=0}^{\infty} (ax)^n = \frac{1}{1-ax}$$

Now, $$\frac{1+(1-4x)^2}{(1-2x)^2(1-4x)} = \frac{A}{1-2x} + \frac{B}{(1-2x)^2} + \frac{C}{1-4x}.$$

$$1+(1-4x)^2 = A(1-2x)(1-4x) + B(1-4x) + C(1-2x)^2$$

After solving $\Rightarrow$

$A = 0$

$B = -2$

$C = 4$

Now $G(x) = 0 - \frac{2}{(1-2x)^2} + \frac{4}{1-4x}$

$$= \frac{4}{1-4x} - \frac{2}{(1-2x)^2}$$

$$= 4(1-4x)^{-1} - 2(1-2x)^{-2}$$

$$G(x) = 4\left[1 + (4x) + (4x)^2 + \cdots + (4x)^n + \cdots\right]$$

$$- 2\left[1 + 2.(2x) + 3(2x)^2 + \cdots + (n+1)(2x)^n + \cdots\right].$$

a_n = Coeff of x^n in $G(x)$

$= 4 \cdot (4)^n - 2(n+1)2^n$

$$\boxed{a_n = 4^{n+1} - (n+1)2^{n+1}}$$

Note: $(1-4x)^{-3} = 1 + 3(4x)^1 + (2)(3)(4x)^2 + (2)(5)(4x)^3 + (3)(5)(4x)^4 + \cdots$

$$= \frac{2}{2}\left[1 + 3(4x)^1 + (2)(3)(4x)^2 + (2)(5)(4x)^3 + (3)(5)(4x)^4 + \cdots\right]$$

$$(1-4x)^{-3} = \frac{1}{2}\left[(1)(2) + (2)(3)(4x) + (4)(3)(4x)^2 + (4)(5)(4x)^3 + (6)(5)(4x)^4 + \cdots + (n+2)(n+1)(4x)^n + \cdots\right].$$

Note: $(4x^2-4x+1)(1-4x)^{-3}$

$$= (4x^2-4x+1)\cdot\frac{1}{2}\Big[(1)(2)+(2)(3)(4x)+(4)(3)(4x)^2 + \cdots + (n)(n-1)(4x)^{n-2} + (n+1)(n)(4x)^{n-1} + (n+2)(n+1)(4x)^n + \cdots\Big]$$

Coeff of x^n: $\frac{1}{2}\left[4(n)(n-1)4^{n-2} - 4(n+1)(n)4^{n-1} + (n+2)(n+1)4^n\right]$

$\Rightarrow \frac{1}{2}\left[n(n-1)4^{n-1} - (n+1)(n)4 + (n+2)(n+1)4^n\right]$.

$\propto$

3. Use the method of generating function to solve the recurrence relation $a_{n+1} - 8a_n + 16a_{n-1} = 4^n;\ n \geq 1$, given that $a_0 = 1$ and $a_1 = 8$.

Sol

$$\sum_{n=1}^{\infty} a_{n+1} x^n = \frac{x}{x} \sum_{n=1}^{\infty} a_{n+1} x^n$$

$$= \frac{1}{x} \sum_{n=1}^{\infty} a_{n+1} x^{n+1}$$

$$= \frac{1}{x} \sum_{n=2}^{\infty} a_n x^n$$

$$= \frac{1}{x} \left[G(x) - a_0 - a_1 x \right] \quad \text{--- (1)}$$

Now,

$$8\sum_{n=1}^{\infty} a_n x^n = 8\left[h(x) - a_0\right]. \quad \text{---(2)}$$

Now

$$16\sum_{n=1}^{\infty} a_{n-1} x^n = 16x \sum_{n=1}^{\infty} a_{n-1} x^{n-1}.$$

$$= 16x \sum_{n=0}^{\infty} a_n x^n = 16x \cdot h(x) \quad \text{---(3)}$$

Now, $\sum_{n=1}^{\infty} 4^n x^n = \sum_{n=1}^{\infty} (4x)^n = \sum_{n=1}^{\infty} (4x)^n + (4x)^0 - (4x)^0$.

$$= \sum_{n=0}^{\infty} (4x)^n - (4x)^0$$

$$= (1-4x)^{-1} - 1.$$

$$= \frac{1}{1-4x} - 1 = \frac{1-(1-4x)}{1-4x}.$$

$$\sum_{n=1}^{\infty} 4^n x^n = \frac{4x}{1-4x}. \quad \text{— (4)}$$

$$n=1$$

$$\sum_{n=1}^{\infty} a_{n+1} x^n - 8\sum_{n=1}^{\infty} a_n x^n + 16\sum_{n=1}^{\infty} a_{n-1} x^n = \sum_{n=1}^{\infty} 4^n x^n$$

$$\frac{1}{x}\left(h(x) - a_0 - a_1 x\right) - 8\left(h(x) - 8a_0\right) + 16x\,h(x) = \frac{4x}{1-4x}.$$

$$h(x) - a_0 - a_1 x - 8x\,h(x) + 8x a_0 + 16x^2 \cdot h(x) = \frac{4x^2}{1-4x}.$$

$$h(x)\left(16x^2 - 8x + 1\right) = \frac{4x^2}{1-4x} + a_0 + a_1 x - 8x a_0.$$

$$G(x).(16x^2 - 8x + 1) = \frac{4x^2}{1-4x} + 1 + 8x - 8x.$$

$$G(x).(1-4x)^2 = \frac{4x^2}{1-4x} + 1$$

$$= \frac{4x^2}{1-4x} + 1.$$

$$G(x) = \frac{4x^2 + 1(1-4x)}{(1-4x)(1-4x)^2}$$

$$= \frac{4x^2 - 4x + 1}{(1-4x)^3}$$

$$= (4x^2 - 4x + 1)(1-4x)^{-3}$$

$$= (4x^2 - 4x + 1)\frac{1}{2}\left[(1)(2) + (2)(3)(4x) + (3)(4)(4x)^2 + \cdots + (n+1)(n+2)(4x)^n + \cdots\right]$$

$$G(x) = (4x^2 - 4x + 1)\frac{1}{2}\Big[(1)(2) + (2)(3)(4x) + (3)(4)(4x)^2 + \cdots + (n-1)(n)(4x)^{n-2} + (n)(n+1)(4x)^{n-1} + (n+1)(n+2)(4x)^n + \cdots\Big].$$

Now,

a_n = Co-eff of x^n in $G(x)$.

$$a_n = \frac{1}{2}\Big[(n-1)(n)4^{n-1} - (n)(n+1)4^n + (n+1)(n+2)4^n\Big].$$

$$= \frac{1}{2}4^{n-1}\Big[n(n-1) + (n)(n+1)4 + (n+1)(n+2)4\Big].$$

$4^n = 4^{n-1} \cdot 4.$

$$= \frac{4^{n-1}}{2}\Big[n^2 - n + 4n^2 + 4n + 4n^2 + 12n + 8\Big]$$

$$a_n = \frac{4^{n-1}}{2}(n^2 + 7n + 8).$$

Module 3

End of the lecture notes!